

Optimsolution: A Cross-Platform Framework for Benchmarking, Sensitivity, and Complexity Analysis of Continuous Optimization Methods

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Abstract

We present Optimsolution, an open-source, cross-platform framework for the systematic benchmarking and analysis of continuous optimization methods, available for Linux and Windows and operable via both a command-line interface (CLI) and a Qt-based graphical user interface (GUI). The framework supports four execution modes: Single mode for single method-problem runs with convergence and distribution analysis; Batch mode for automated multi-method, multi-problem experimentation with aggregated statistical summaries; method sensitivity analysis for quantifying the effect of algorithmic hyper-parameters on solution quality; and problem sensitivity analysis for assessing how problem-defining parameters influence instance difficulty and inter-method rankings. Complexity analysis is provided along two axes method scalability across problem dimensionalities, and problem difficulty profiles across method portfolios. Statistical validation is embedded natively through Wilcoxon signed-rank and Friedman tests with post-hoc pairwise analysis, complemented by rank plots and box-plot visualisations. All outputs are exportable as CSV files or publication-ready PNG figures. The framework is designed for extensibility: new methods and benchmark problems can be registered and activated through the GUI without modifications to the core codebase, supporting rapid experimental iteration.

Keywords: Continuous optimization; Benchmarking framework; Metaheuristics; Sensitivity analysis; Complexity analysis; GUI; CLI; cross-platform

Charilogis, V., Tsoulos, I.G., Gianni, A. M.

Received:

Revised:

Accepted:

Published:

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